Observer Quality as a Resource Constraint in Quantum Darwinism: Optimal Decoding, ε-Approximate Spectrum Broadcast Structure, and a Central-Spin Worked Example

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February 7, 2026

Abstract

Quantum Darwinism (QD) and Spectrum Broadcast Structure (SBS) formalize a mechanism by which many observers can learn the same classical pointer value of a system by measuring different fragments of its environment. Most QD analyses treat the observer as idealized: unlimited access, perfect calibration, and no temporal constraints. This paper introduces an explicit *observer quality* parameterization and propagates it through redundancy and objectivity statements in a way that remains stable under an explicit ε-SBS trace-distance error model.

Our main technical contributions are: (i) a decoder-level, tight (Chernoff-optimal) sample-complexity characterization for the number of environmental fragments needed to infer a pointer value under a calibrated observer model, (ii) a data-processing theorem showing that calibration noise (modeled as a CPTP channel) degrades the quantum Chernoff exponent, and (iii) an ε-robustness theorem upgrading ideal-SBS sample-complexity bounds to approximate-SBS states with additive error control.

To address "no concrete physics" concerns, we provide a fully worked open-system example: a central-spin pure-dephasing model whose conditional fragment states can be computed analytically, yielding explicit formulas for the observer parameters (R_O, C_O, τ_O) as functions of couplings, readout noise, and acquisition time. We also highlight a sharp performance gap between collective (coherent) decoding and *local* (Helstrom-per-fragment) product decoding for pure-state records, directly linking *quantum memory horizon* to redundancy. Finally, to connect this physics formalism to the DLN observer-quality thesis, we introduce a three-state stage index $q_O \in \{q_{\mathrm{D}}, q_{\mathrm{L}}, q_{\mathrm{N}}\}$ mapping Dot/Linear/Network stages to memoryless, local-product, and collective decoding classes.

We then prove three DLN-enabled dynamical results about *observer topology* over time: (A) a dynamical redundancy theorem showing that the long-run effective Chernoff exponent depends on the observer's revision topology $\mathcal{R}_O$, with full-cycle revision achieving the adaptive optimum while expand-only revision collapses permanently to the linear baseline; (B) an "inverted sophistication" theorem showing that an *unmonitored* collective decoder can be strictly worse than optimal product decoding below a critical coherence fraction; and (C) a pointer-accessibility proposition formalizing how the DLN stage determines which pointer distinctions are resolvable from a fixed fragment budget, providing an operational basis for observer-designed control of accessible quantum outcomes.

1 Introduction

Quantum Darwinism proposes that classical objectivity arises when information about a preferred pointer observable of a system is redundantly encoded in the environment [? ? ?]. A complementary structural characterization is Spectrum Broadcast Structure [? ?], and strong-QD variants connect entropic and geometric criteria [? ?].

A persistent modeling gap is that *observers are usually treated as ideal*. Yet in realistic experiments or cognitive settings, an observer may: (i) access only a fraction of environment fragments, (ii) have imperfect measurement calibration, and (iii) be limited by a temporal integration horizon (finite memory or pointer stability). There is substantial prior work on non-idealities in the *environment* (e.g., "hazy" environments [?]) and on generic emergence of objectivity [? ?], but the observer itself is rarely parameterized at the same level of explicitness.

Scope of this paper. We restrict to a single core task: *observer-relative redundancy/objectivity statements* in QD/SBS when the observer is explicitly modeled by a quality triple

$$Q_O = (R_O, \Lambda_O, \tau_O),$$

where R_O is an access fraction, Λ_O is a calibration channel applied to each accessed fragment prior to readout, and τ_O is an acquisition or memory horizon. For interpretability (and to align with existing QD language) we also extract a scalar *calibration index* C_O from Λ_O in the binary case.

Why an ε-SBS error model. Many QD/SBS derivations assume conditional i.i.d. record models. Instead, we start from an explicit trace-distance approximation: the actual joint state $\rho_{S\mathcal{E}}$ is within ε in trace distance of an *ideal* SBS state $\sigma_{S\mathcal{E}}$. We then push sample-complexity statements through this approximation using data processing and decision-theoretic continuity bounds.

Observer topology over time: DLN-enabled dynamical predictions. The quality triple Q_O captures *static* limitations within a single observational episode. However, in realistic QD/SBS protocols the observation model may be non-stationary (e.g., coherence loss across the fragment register, drifting calibration, or regime switching in the environment), so the observer's capacity to *revise* its decoding assumptions becomes a second, structurally distinct resource. Motivated by the DLN revision-graph formalism, we introduce an observer-specific revision topology $\mathcal{R}_O$ over decoder models and establish how this topology governs the long-run redundancy rate. This yields three results (Sec. 9): dynamical redundancy as a topological invariant of $\mathcal{R}_O$ (Result A), inverted sophistication for unmonitored collective decoding (Result B), and pointer accessibility as a stage-dependent resolution bound (Result C).

2 Related work and positioning

This paper builds on three strands:

1. **QD and SBS foundations.** The environment-as-witness framing originates in ?] and is synthesized by ? ?]. SBS provides a structural notion of objectivity [?], and a unifying review is ?]. Strong-QD links entropic and structural conditions [?].

2. **Non-ideal environments and fragment size.** "Hazy" environments reduce redundancy and record capacity [?]. Photon-scattering models exhibit huge redundancy in everyday environments [? ?].

3. **Hypothesis testing and optimal decoding.** Binary quantum hypothesis testing is classical material [? ?]. The quantum Chernoff bound characterizes the optimal asymptotic error exponent [? ?]. These tools are natural for QD because "learning the pointer value" is precisely a discrimination task between conditional fragment states.

We also connect to recent resource-theoretic and operational approaches emphasizing *accessible information* rather than mutual information [?].

3 Setup: SBS, ε-SBS, and observer quality

3.1 Spectrum Broadcast Structure and its approximation

Let S be the system and $\mathcal{E} = \mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_N$ a decomposition of the environment into fragments. We focus on a classical pointer random variable X taking values in a finite alphabet $\mathcal{X}$ (often $\mathcal{X} = \{0, 1\}$).

Definition 1 (SBS state)**.** A joint state $\sigma_{S\mathcal{E}}$ has *spectrum broadcast structure* (SBS) with respect to pointer basis $\{|x\rangle\}_{x\in\mathcal{X}}$ if it can be written as

$$\sigma_{S\mathcal{E}} = \sum_{x\in\mathcal{X}} p_x \, |x\rangle\langle x|_S \otimes \bigotimes_{k=1}^{N} \sigma_{\mathcal{E}_k}^{(x)}, \tag{1}$$

where p is a probability distribution and, for each fragment k, the conditional states $\{\sigma_{\mathcal{E}_k}^{(x)}\}_{x\in\mathcal{X}}$ have mutually orthogonal supports (perfect distinguishability) [? ?].

Definition 2 (ε-SBS)**.** A state $\rho_{S\mathcal{E}}$ is *ε-SBS* if there exists an SBS state $\sigma_{S\mathcal{E}}$ such that

$$D_{\mathrm{tr}}(\rho_{S\mathcal{E}}, \sigma_{S\mathcal{E}}) \le \varepsilon, \qquad D_{\mathrm{tr}}(\rho, \sigma) := \tfrac{1}{2} \left\| \rho - \sigma \right\|_1 .$$

3.2 Observer quality

An observer interacts only with a subset of fragments and through an imperfect instrument.

Definition 3 (Observer quality)**.** An observer is specified by a quality triple

$$Q_O = (R_O, \Lambda_O, \tau_O),$$

where:

1. $R_O \in [0, 1]$ is an *access fraction*: out of N fragments, O can access at most $m_{\max} = \lfloor R_O N \rfloor$ fragments in a given observational episode;
2. Λ_O is a CPTP map acting on each accessed fragment, representing calibration noise, coarse-graining, or any pre-measurement processing;
3. τ_O is a temporal horizon (memory, stability, or acquisition time). In the central-spin model (Sec. 7), it is mapped to an acquisition time budget and a pointer-stability constraint.

When $\mathcal{X} = \{0, 1\}$, we can compress Λ_O to a scalar "single-fragment calibration" index.

Definition 4 (Binary calibration index)**.** Let ρ_0, ρ_1 be the two conditional states of a single accessed fragment after Λ_O, and assume equal priors. Define

$$C_O := 1 - P_e^\star(\rho_0, \rho_1), \qquad P_e^\star(\rho_0, \rho_1) = \tfrac{1}{2}\left(1 - \tfrac{1}{2}\left\|\rho_0 - \rho_1\right\|_1\right), \tag{2}$$

where $P_e^\star$ is the Helstrom-optimal single-copy Bayes error [? ?]. Then $C_O \in [\frac{1}{2}, 1]$.

3.3 DLN stage index and three discrete observer-quality states

The resource triple $Q_O = (R_O, \Lambda_O, \tau_O)$ is a *continuous* parameterization of observer limitations: it specifies how many fragments can be accessed, how each accessed record is distorted by calibration, and how long the observer can integrate across fragments. For several questions in quantum Darwinism it is also useful to coarse-grain observers into a small number of *information-processing regimes*. Motivated by the Dot–Linear–Network (DLN) observer-quality framework, which formalizes stage differences via belief-dependency graphs and a revision graph over model space [**?**], we introduce a three-valued stage variable

$$q_O \in \{q_{\mathrm{D}}, q_{\mathrm{L}}, q_{\mathrm{N}}\} \equiv \{0, 1, 2\}.$$

In this paper, q_O is defined operationally by the class of admissible decoding procedures on an m-fragment register within the observer's temporal horizon.

Definition 5 (Three q-states as decoder classes)**.** Fix a fragment set $A \subseteq \{1, \ldots, N\}$ with $|A| = m$ and conditional states $\rho_A^{(x)} := \bigotimes_{k \in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(x)})$ (SBS witness experiment, $\mathcal{X} = \{0, 1\}$). A *decoder* is a measurement–decision procedure producing an estimate $\widehat{X} \in \{0, 1\}$ from the m-fragment quantum register. We define three nested decoder classes:

(i) q_{D} **(Dot).** *Single-fragment / memoryless decoding.* The decoder acts on at most one fragment at a time and retains no state across fragments. Equivalently, the effective hypothesis test is restricted to a single-copy test on $\Lambda_O(\sigma_{\mathcal{E}_k}^{(x)})$ for some k.

(ii) q_{L} **(Linear).** *Product decoding with classical postprocessing.*

(iii) q_{L} **(Linear).** *Local (Helstrom-per-fragment) decoding with classical postprocessing.* The decoder measures each fragment immediately with the (equal-prior) Helstrom POVM optimized for single-fragment discrimination between $\Lambda_O(\sigma_{\mathcal{E}_k}^{(0)})$ and $\Lambda_O(\sigma_{\mathcal{E}_k}^{(1)})$, producing a classical outcome per fragment. It then applies an arbitrary classical decision rule to the m outcomes. This class captures "branch-by-branch" processing in which each fragment contributes evidence independently and no inter-fragment quantum coherence is maintained.

(iv) q_{N} **(Network).** *Collective decoding.* The decoder may apply an arbitrary POVM on the joint register $\bigotimes_{k \in A} \mathcal{E}_k$ (equivalently, any collective m-copy hypothesis test), attaining Chernoff-optimal error exponents when the record model is i.i.d.

Remark 1 (Relation to finite-resource sample complexity)**.** The results below quantify how the required fragment count m scales with the error target δ and with calibration noise. This is a finite-resource regime in the sense of non-asymptotic quantum information theory [**?** **?**]. Recent work has sharpened the *sample-complexity* viewpoint in quantum hypothesis testing and related tasks [**?**]. In our setting, the relevant operational distinction is whether $q_O = q_{\mathrm{L}}$ or $q_O = q_{\mathrm{N}}$: under q_{N} collective decoding achieves the quantum Chernoff exponent, while under q_{L} (local Helstrom-per-fragment decoding) the achievable exponent can be strictly smaller in physically relevant regimes (Cor. 1).

Table 1: Mapping between three observer-quality q-states and DLN stages. The DLN column follows the graph-theoretic stage definitions in **?**]. The decoding column specifies the admissible measurement class on m accessed fragments in this paper.

q_O	DLN stage	DLN representational signature	Operational signature (QD decoding)
q_{D}	Dot	No persistent belief-dependency graph ($G = \emptyset$); reactive policy.	Single-fragment or memoryless decoding; no $m > 1$ aggregation.
q_{L}	Linear	Null belief graph over evidence nodes; no cross-node information propagation.	Local (Helstrom-per-fragment) measurements with classical postprocessing; no coherent joint decoding across fragments.
q_{N}	Network	Shared latent structure connecting evidence nodes (factor reuse) and, in the full DLN instantiation, a model-revision cycle over $\mathcal{M}$.	Collective decoding on $\bigotimes_{k\in A}\mathcal{E}_k$; Chernoff-optimal exponents are feasible when m-copy coherence is available.

4 Optimal decoding and Chernoff sample complexity

4.1 Binary hypothesis testing on fragments

Fix an SBS reference state $\sigma_{S\mathcal{E}}$ of the form (1) with $\mathcal{X} = \{0, 1\}$. For any fragment set $A \subseteq \{1, \ldots, N\}$, let

$$\sigma_A^{(x)} := \bigotimes_{k\in A} \sigma_{\mathcal{E}_k}^{(x)}$$

denote the conditional state of the accessible environment.

The observer first applies Λ_O to each accessed fragment and then performs an optimal measurement to decide x. This induces a binary quantum hypothesis testing problem:

$$H_0 : \ \rho_A = \bigotimes_{k\in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(0)}), \qquad H_1 : \ \rho_A = \bigotimes_{k\in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(1)}).$$

4.2 Quantum Chernoff bound and tight redundancy scaling

Define the (binary) quantum Chernoff coefficient for a pair of states ρ, σ:

$$Q(\rho, \sigma) := \min_{s\in[0,1]} \mathrm{Tr}\left[\rho^s \sigma^{1-s}\right],$$

and the associated Chernoff exponent

$$\xi(\rho, \sigma) := -\log Q(\rho, \sigma).$$

The quantum Chernoff bound states that for discriminating $\rho^{\otimes n}$ vs $\sigma^{\otimes n}$ with equal priors, the optimal error decays with exponent $\xi(\rho, \sigma)$ [**?** **?**].

We use a finite-n inequality that is convenient for sample-complexity statements.

Theorem 1 (Finite-n Chernoff upper bound)**.** *Let ρ, σ be density operators and consider equal priors on $H_0 : \rho$ and $H_1 : \sigma$. Then the optimal Bayes error obeys*

$$P_e^\star(\rho, \sigma) \leq \tfrac{1}{2}\, Q(\rho, \sigma). \tag{3}$$

Consequently, for product states over a fragment set A,

$$P_e^\star\left(\bigotimes_{k\in A}\rho_k, \bigotimes_{k\in A}\sigma_k\right) \le \tfrac{1}{2}\min_{s\in[0,1]}\prod_{k\in A}\mathrm{Tr}\left[\rho_k^s\sigma_k^{1-s}\right].$$

Proof. Inequality (3) is a standard consequence of the Audenaert et al. bound and is commonly stated as $P_e^\star(\rho,\sigma) \le \frac{1}{2}\min_{s\in[0,1]}\mathrm{Tr}[\rho^s\sigma^{1-s}]$ [**?**]. The product form follows from multiplicativity of the trace under tensor products. □

Definition 6 (Observer-effective Chernoff exponent on a fragment set)**.** For an observer O and fragment set A, define

$$\xi_O(A) := \max_{s\in[0,1]}\left(-\sum_{k\in A}\log\mathrm{Tr}\left[\Lambda_O(\sigma_{\mathcal{E}_k}^{(0)})^s\,\Lambda_O(\sigma_{\mathcal{E}_k}^{(1)})^{1-s}\right]\right). \tag{4}$$

Proposition 1 (Access threshold as an information budget)**.** *Fix an error target $\delta \in (0,\frac{1}{2})$ and an SBS reference experiment. If an observer can access at most $m_{\max} = \lfloor R_O N\rfloor$ fragments, then a sufficient condition for δ-decoding of the pointer value is the existence of a set A with $|A| \le m_{\max}$ such that*

$$\xi_O(A) \ge \log\left(\frac{1}{2\delta}\right).$$

Proof. By Theorem 1, $P_e^\star(A) \le \frac{1}{2}\exp(-\xi_O(A))$. If $\xi_O(A) \ge \log(1/(2\delta))$, then $P_e^\star(A) \le \delta$. □

4.3 Calibration as distinguishability contraction

A central point for "non-ideal observers" is that calibration noise cannot improve distinguishability. For the Chernoff coefficient, this is a data-processing statement.

Theorem 2 (Calibration degrades Chernoff distinguishability)**.** *Let Λ be a CPTP map and ρ,σ quantum states. For every $s \in [0,1]$,*

$$\mathrm{Tr}\left[\Lambda(\rho)^s\Lambda(\sigma)^{1-s}\right] \;\ge\; \mathrm{Tr}\left[\rho^s\sigma^{1-s}\right].$$

Equivalently, $Q(\Lambda(\rho),\Lambda(\sigma)) \ge Q(\rho,\sigma)$ and therefore

$$\xi(\Lambda(\rho),\Lambda(\sigma)) \le \xi(\rho,\sigma).$$

Proof. For $s \in (0,1)$, define the Petz-Rényi divergence $D_s(\rho\|\sigma) = \frac{1}{s-1}\log\mathrm{Tr}[\rho^s\sigma^{1-s}]$. For $s \in (0,1)$ this divergence satisfies the data-processing inequality $D_s(\Lambda(\rho)\|\Lambda(\sigma)) \le D_s(\rho\|\sigma)$ under CPTP maps [**? ? ?**]. Rearranging yields the claimed inequality for $\mathrm{Tr}[\cdot]$. The endpoint cases $s = 0,1$ follow by continuity. □

5 ε-SBS robustness of decision rules

The point of an explicit trace-distance SBS approximation is that it immediately controls *any* decision procedure via data processing.

Lemma 1 (Decision-theoretic continuity under trace distance)**.** *Let ρ and σ be two states on the same Hilbert space and let* Dec *be any (possibly adaptive) measurement-plus-decision procedure outputting a discrete hypothesis $\widehat{X}$. Then*

$$\left|\mathbb{P}_\rho(\widehat{X} \ne X) - \mathbb{P}_\sigma(\widehat{X} \ne X)\right| \le D_{\mathrm{tr}}(\rho,\sigma).$$

Proof. A measurement-plus-decision procedure is a CPTP map from states to a classical distribution on $\widehat{X}$. Trace distance contracts under CPTP maps, so the total variation distance between the induced classical distributions is bounded by $D_{\mathrm{tr}}(\rho, \sigma)$ [?]. The error event $\{\widehat{X} \neq X\}$ is a measurable subset of outcomes, hence its probability differs by at most the total variation distance. □

Theorem 3 (ε-robust Chernoff sample complexity)**.** *Let $\rho_{S\mathcal{E}}$ be ε-SBS with witness $\sigma_{S\mathcal{E}}$. Fix a fragment set A and an observer O. Let $\delta \in (0, \frac{1}{2})$ be a target error under the* actual *state ρ.*

If $\xi_O(A) \geq \log\left(\frac{1}{2(\delta-\varepsilon)}\right)$ and $\delta > \varepsilon$, then there exists a decision procedure on A (namely the optimal decoder for the witness experiment) such that

$$\mathbb{P}_\rho(\widehat{X} \neq X) \leq \delta.$$

Proof. Apply Proposition 1 to the witness state σ: with the given $\xi_O(A)$, there exists a decoder achieving $\mathbb{P}_\sigma(\widehat{X} \neq X) \leq \delta - \varepsilon$. Then Lemma 1 gives $\mathbb{P}_\rho(\widehat{X} \neq X) \leq (\delta - \varepsilon) + \varepsilon = \delta$. □

6 Local versus collective decoding: a sharp gap for pure records

Observer temporal limitations often restrict *which measurements are feasible.* A key operational distinction is between (i) collective measurements across m fragments (requiring quantum memory to preserve coherence across the m-fragment register) and (ii) *local* decoding that measures each fragment immediately.

For two pure states $|\psi_0\rangle, |\psi_1\rangle$ with overlap

$$c := |\langle\psi_0\rangle \psi_1| \in (0, 1),$$

the m-copy conditional states are $|\psi_0\rangle^{\otimes m}$ and $|\psi_1\rangle^{\otimes m}$ with overlap c^m. The quantum Chernoff coefficient is c^{2m}, so the collective (Chernoff-optimal) error exponent is $-\log(c^2)$ per copy [? ?].

For the q_{L} class in Definition 5, we consider local decoding in which *each fragment is measured with the equal-prior Helstrom POVM for the single-fragment pair.* For pure states, there exists a POVM attaining the fidelity bound in the sense that the induced classical distributions p, q satisfy

$$\sum_y \sqrt{p(y)q(y)} = c$$

[? ?]. For the resulting symmetric binary distributions, the classical Chernoff coefficient equals this Bhattacharyya coefficient (attained at $s = \frac{1}{2}$), hence the local Helstrom-per-fragment error exponent is $-\log c$ per copy.

Corollary 1 (Exact factor-of-two exponent gap for pure-state records under local Helstrom decoding)**.** *Let $|\psi_0\rangle, |\psi_1\rangle$ be pure states with overlap $c \in (0, 1)$. Let $P_e^{\mathrm{coll}}(m)$ be the optimal Bayes error using* collective *measurements on m copies, and let $P_e^{\mathrm{L}}(m)$ be the Bayes error achieved by applying the single-copy Helstrom POVM to each copy independently and then performing optimal classical postprocessing on the m outcomes. Then the asymptotic error exponents satisfy*

$$\lim_{m\to\infty} -\frac{1}{m} \log P_e^{\mathrm{coll}}(m) = -\log(c^2), \qquad \lim_{m\to\infty} -\frac{1}{m} \log P_e^{\mathrm{L}}(m) = -\log(c).$$

In particular, the local Helstrom-per-fragment exponent is exactly half the collective-measurement exponent.

Proof. For pure states, the quantum Chernoff coefficient equals the fidelity, so $Q(\psi_0, \psi_1) = c^2$ and the collective Chernoff exponent is $-\log(c^2)$ [? ?].

For the local Helstrom-per-fragment measurement, the induced single-copy outcome distributions are symmetric: under H_0 the correct-decision outcome occurs with probability $1 - P_e^\star$ and under H_1 with probability $P_e^\star$, and vice versa, where $P_e^\star$ is the equal-prior Helstrom error. For this symmetric binary pair, the Bhattacharyya coefficient equals $2\sqrt{P_e^\star(1 - P_e^\star)}$, which equals c for pure states. For i.i.d. classical samples, the Chernoff coefficient of the m-sample distributions is the mth power of the single-sample coefficient, hence equals c^m, yielding exponent $-\log c$. □

Remark 2 (What this gap does and does not claim)**.** The strict factor-of-two separation in Cor. 1 is for the *local Helstrom-per-fragment* q_{L} class used throughout this paper. It is not a claim that *all* product-measurement strategies are separated from collective measurements for pure-state discrimination. If an observer can redesign its local POVM jointly across copies (still without coherent quantum memory), the optimal product-measurement exponent can approach the quantum Chernoff exponent in some settings. Our restriction to local Helstrom-per-fragment decoding is deliberate: it models an experimentally and cognitively natural regime in which each fragment is processed with a fixed, single-fragment-calibrated instrument and only classical aggregation is available.

7 Worked physical model: central-spin pure dephasing

We now provide a concrete open-system model and compute (R_O, C_O, τ_O) in terms of physical parameters.

7.1 Model

Let S be a qubit with pointer basis the eigenbasis of σ_z^S. Let the environment consist of N qubits with $\sigma_z^{(k)}$. Consider the pure-dephasing Hamiltonian

$$H = \sigma_z^S \otimes \sum_{k=1}^{N} g_k \sigma_z^{(k)}, \tag{5}$$

and initial environment state $|+\rangle^{\otimes N}$ with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Conditioned on the pointer value $x \in \{0, 1\}$ (eigenvalues ± 1 of σ_z^S), fragment k evolves to a pure state

$$|\phi_k^{(x)}(t)\rangle = e^{\mp i g_k t \sigma_z} |+\rangle ,$$

so that the single-fragment overlap is

$$c_k(t) := \left| \langle \phi_k^{(0)}(t)\rangle \, \phi_k^{(1)}(t) \right| = |\cos(2 g_k t)| . \tag{6}$$

This model (and its variants) is standard in decoherence theory and is directly connected to SBS formation in spin environments [? ?].

7.2 Exact Chernoff exponent and fragment requirement

For pure states, the Chernoff coefficient equals the fidelity:

$$Q\left(|\phi_k^{(0)}(t)\rangle , |\phi_k^{(1)}(t)\rangle\right) = c_k(t)^2, \qquad \xi_k(t) = -\log c_k(t)^2.$$

For a set A of accessed qubits,

$$\xi(A;t) = \sum_{k\in A} -\log\left(\cos^2(2g_k t)\right), \tag{7}$$

and Theorem 1 implies

$$P_e^\star(A;t) \le \tfrac{1}{2}\exp(-\xi(A;t)).$$

Thus a sufficient condition for error $\le \delta$ is $\xi(A;t) \ge \log(1/(2\delta))$.

7.3 Mapping to observer parameters

Access fraction R_O. In a laboratory implementation, an observer may only intercept a fraction of environment qubits (geometric coverage, detector efficiency, spatial range). If $m_{\max}$ qubits are accessible per trial out of N, then $R_O = m_{\max}/N$.

Calibration C_O. Suppose the observer's instrument applies a depolarizing channel on each accessed environment qubit: $\Lambda_p(\rho) = (1-p)\rho + p\frac{\mathbf{1}}{2}$, capturing miscalibration or noise before readout. For any two qubit states, depolarization contracts trace distance by a factor $(1-p)$ [**?**]. For the central-spin conditional pure states, the single-copy Helstrom success probability becomes

$$C_O(t) = \tfrac{1}{2}\left(1 + (1-p)\sqrt{1 - c_k(t)^2}\right) = \tfrac{1}{2}\left(1 + (1-p)|\sin(2g_k t)|\right),$$

where we used $c_k(t) = |\cos(2g_k t)|$. If couplings differ, $C_O(t)$ varies with k and can be summarized by a worst-case or average calibration across accessible fragments.

Temporal horizon τ_O. If the observer performs collective decoding across m fragments, a quantum memory must preserve coherence for at least the acquisition time of those m qubits plus the decoding time. Let t_{meas} be the per-qubit acquisition time. A minimal feasibility constraint is $\tau_O \gtrsim m\, t_{\text{meas}}$. Additionally, the pointer value must remain stable over the acquisition interval, typically limited by system relaxation times (e.g., T_1) [**? ?**]. In this model, we treat stability as an explicit constraint rather than imposing an unjustified "exponential stability" assumption.

7.4 Numerical illustration and reproducibility

The accompanying repository includes a script that samples couplings $\{g_k\}$, computes $\xi(A;t)$ for the best accessible subset under an access fraction R_O, and plots redundancy as a function of time t. This is included to make the "worked example" fully reproducible. Figure 1 shows a typical redundancy profile as a function of interaction time in this toy model.

8 DLN-series connection and stage mapping

This paper can be read independently as a finite-resource analysis of observer-limited decoding in the QD/SBS setting. Nevertheless, it is motivated by a broader program in which "observer quality" is treated as a property of the *information-processing topology* used to integrate evidence. In the Dot–Linear–Network (DLN) compression model, stage differences are formalized by two objects: (i) a *belief-dependency graph* G governing how evidence updates propagate at the inference level, and (ii) a *revision graph* $\mathcal{R}$ over a model space $\mathcal{M}$ governing when an observer can revise its representational assumptions [**?**].

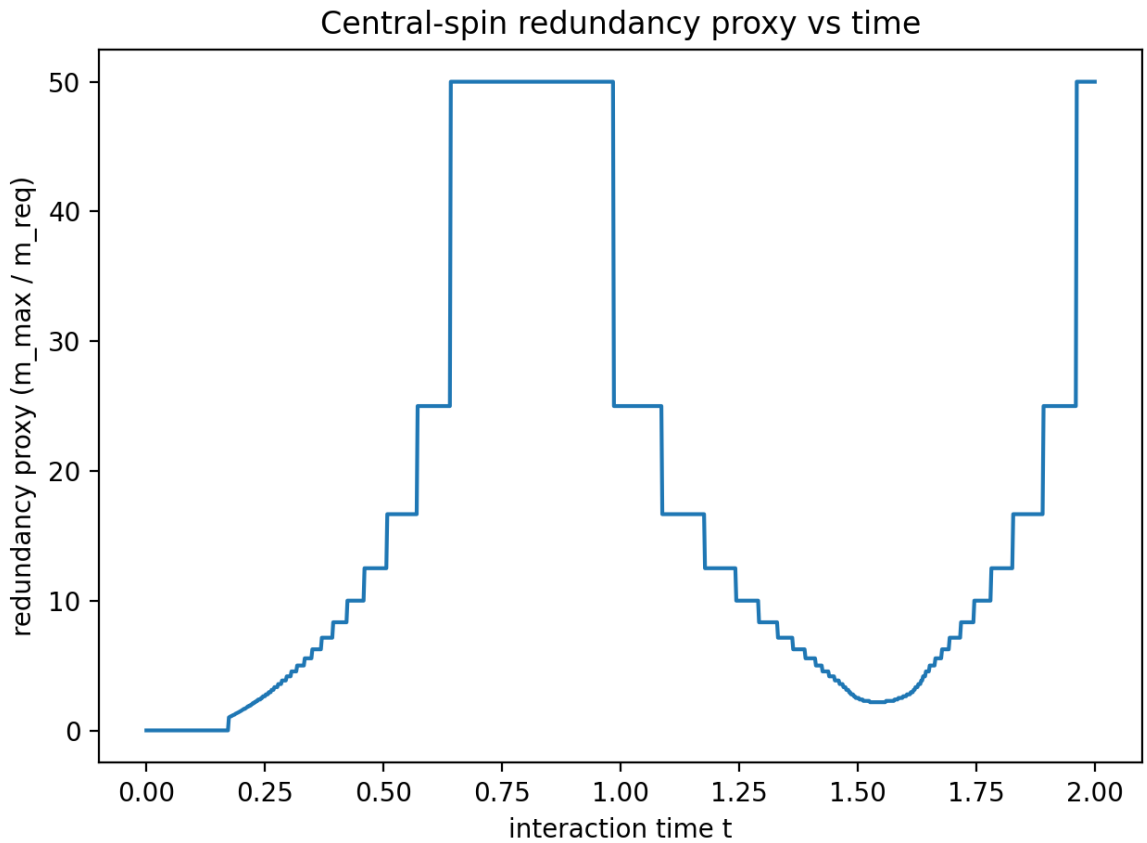

Figure 1: Central-spin toy model (Sec. 7): redundancy (accessible copies) versus interaction time t for a sampled distribution of couplings $\{g_k\}$, access fraction $R_O = 0.25$, and target decoding error $\delta = 10^{-3}$. The curve is produced by the reproducibility script in the repository.

The three q-states introduced in Sec. 3.3 provide an operational instantiation of these stages for quantum Darwinism: q_{D} corresponds to memoryless single-fragment decoding (no persistent dependency structure), q_{L} corresponds to local (Helstrom-per-fragment) decoding with classical aggregation (independent evidence nodes), and q_{N} corresponds to collective decoding that exploits the shared latent pointer record as an explicit common factor (Table 1). In this physics setting, the resource triple $Q_O = (R_O, \Lambda_O, \tau_O)$ determines which q-state is feasible by constraining access, calibration, and the time available to maintain coherence across fragments.

We emphasize that DLN is used here as a representational classification and a source of modeling priors for bounded observers. The quantitative claims of this paper (Chernoff-based sample complexity, calibration contraction, and ε-robustness) are self-contained and do not presuppose any developmental interpretation. Rather than deferring structural revision, we incorporate the DLN revision graph directly: we define an observer-specific revision topology $\mathcal{R}_O$ over admissible decoder models and show that its topological features (notably the presence or absence of a directed cycle) control dynamical redundancy and pointer accessibility in non-stationary QD/SBS protocols (Sec. 9).

9 Observer topology and three new results

The preceding sections treat observer quality as a *within-episode* resource constraint: for a fixed SBS witness experiment, the observer chooses a decoder class (Sec. 3.3) and incurs sample complexity governed by a Chernoff-type exponent that contracts under calibration noise and remains stable under ε-SBS perturbations. Here we add the DLN second layer: a *revision topology* that constrains how an observer can revise its decoding assumptions across time in a non-stationary observation model. This yields three results that are not visible in a purely static analysis.

9.1 Non-stationary regimes, revision topology, and monitoring

We model an experiment as a sequence of episodes $t = 1, 2, \ldots$. In episode t the observer attempts to infer a pointer variable $X_t \in \mathcal{X}$ from an m-fragment register A_t. The conditional fragment states may vary over time due to drift, coherence loss, or regime switching. We capture this by an unobserved regime parameter $\theta_t \in \Theta$, with conditional joint states $\rho_{A_t}^{(x)}(\theta_t)$. The observer may deploy a decoder from a class $q \in \{q_{\mathrm{D}}, q_{\mathrm{L}}, q_{\mathrm{N}}\}$, but in general the *appropriate* decoder depends on θ_t.

Definition 7 (Observer revision graph)**.** Let $\mathcal{M}$ be a set of decoder models (or decoding assumptions) available to an observer, e.g., $\mathcal{M} = \{\mathrm{Dec}_{\mathrm{L}}, \mathrm{Dec}_{\mathrm{N}}\}$ for product versus collective decoding. The *observer revision graph* is a directed graph

$$\mathcal{R}_O = (\mathcal{M}, \mathcal{T}_O), \qquad \mathcal{T}_O \subseteq \mathcal{M} \times \mathcal{M},$$

whose edges specify which model switches the observer can execute between episodes. We call $\mathcal{R}_O$ *expand-only* if $\mathcal{R}_O$ is acyclic (no directed cycle), and *full-cycle* if $\mathcal{R}_O$ contains a directed cycle.

In the DLN framework, $\mathrm{Dec}_{\mathrm{N}}$ corresponds to a factorized representation in which fragments share a latent pointer record, while $\mathrm{Dec}_{\mathrm{L}}$ corresponds to an expanded representation in which fragments are treated independently. An expansion transition $\mathrm{Dec}_{\mathrm{N}} \to \mathrm{Dec}_{\mathrm{L}}$ is triggered by detected failure of the shared-latent assumption; a contraction transition $\mathrm{Dec}_{\mathrm{L}} \to \mathrm{Dec}_{\mathrm{N}}$ is triggered when diagnostic evidence supports restoring coherent joint processing.

Definition 8 (Monitoring action and monitoring cost)**.** A *monitoring action* is any procedure that produces an estimate $\widehat{\theta}_t$ of the current regime parameter θ_t with misclassification probability at most η_t. We model monitoring as consuming an episode budget fraction $C_{\mathrm{monitor}} \in (0, 1)$, leaving an effective decoding budget fraction $1 - C_{\mathrm{monitor}}$ for inferring X_t. Operationally, C_{monitor} can represent: (i) a fraction of fragments reserved for diagnostic tests, (ii) an acquisition-time overhead that reduces usable coherence window, or (iii) calibration work that reduces integration time.

Principle 1 (Monitoring is never free)**.** *In non-stationary observation models, any policy that keeps the probability of applying a regime-inappropriate decoder exponentially small in the fragment budget necessarily incurs a nonzero monitoring cost* $C_{\mathrm{monitor}} > 0$.

Remark 3 (Why Principle 1 is operationally unavoidable)**.** To adapt between regimes with exponentially small failure probability, the observer must distinguish regimes with exponentially small misclassification probability. If each diagnostic fragment yields a positive Chernoff information $\xi_{\mathrm{diag}} > 0$ for testing $\theta \in \{\theta^{\mathrm{good}}, \theta^{\mathrm{bad}}\}$, then achieving $\eta_t \leq e^{-\beta m}$ requires at least $m_{\mathrm{diag}} \geq (\beta/\xi_{\mathrm{diag}})m$ diagnostic fragments. Thus $C_{\mathrm{monitor}} = m_{\mathrm{diag}}/m \geq \beta/\xi_{\mathrm{diag}} > 0$. The value of C_{monitor} is a physical parameter determined by the diagnostic channel and the reliability requirement.

9.2 Result A: dynamical redundancy depends on revision topology

We quantify long-run performance using a time-averaged effective error exponent. Let $P_{e,t}$ be the episode-t decoding error for inferring X_t. Define the effective exponent

$$\xi_{\text{eff}} := \liminf_{T\to\infty} \left(-\frac{1}{Tm} \sum_{t=1}^{T} \log(2P_{e,t}) \right). \tag{8}$$

Remark 4 (Ensemble-average exponent and Jensen gap)**.** In addition to ξ_{eff} (a "typical" or "quenched" exponent defined by the average of $\log P_{e,t}$), it is often operationally relevant to track the exponent of the *average* error probability. Define the ensemble exponent

$$\xi_{\text{ens}} := \liminf_{T\to\infty} \left(-\frac{1}{m} \log\left(\frac{2}{T} \sum_{t=1}^{T} P_{e,t} \right) \right).$$

For i.i.d. episodes, $\xi_{\text{ens}} = -\frac{1}{m}\log\big(2\,\mathbb{E}[P_{e,t}]\big)$. By Jensen's inequality, $\xi_{\text{ens}} \le \xi_{\text{eff}}$. In the numerical verification below (Sec. 9.6), we report ξ_{ens} because it captures "rare but catastrophic" unmonitored failures that set an error floor.

For a fixed regime θ, define the best achievable *per-fragment* exponents within decoder classes by

$$\xi_{\text{L}}(\theta) := \limsup_{m\to\infty} -\frac{1}{m}\log\big(2P^{\star}_{e,\text{L}}(m;\theta)\big), \qquad \xi_{\text{N}}(\theta) := \lim_{m\to\infty} -\frac{1}{m}\log\big(2P^{\star}_{e,\text{N}}(m;\theta)\big),$$

where $P^{\star}_{e,\text{L}}$ is the optimal error under local Helstrom-per-fragment decoding (class q_{L}) and $P^{\star}_{e,\text{N}}$ is the optimal error under collective measurements (class q_{N}) when feasible.

Theorem 4 (Dynamical redundancy as a topological invariant of $\mathcal{R}_O$)**.** *Assume a two-regime model $\theta_t \in \{\theta^{\text{good}}, \theta^{\text{bad}}\}$ with $\mathbb{P}(\theta_t = \theta^{\text{good}}) = f \in (0,1)$ i.i.d. over episodes. Assume that in the good regime collective decoding strictly improves the exponent, $\xi_{\text{N}}(\theta^{\text{good}}) > \xi_{\text{L}}(\theta^{\text{good}})$, while in the bad regime product decoding is robust and dominates any fixed, unmonitored collective decoder in the sense that there exists a "stale" collective decoder $\text{Dec}_{\text{N}}^{\text{stale}}$ with exponent $\xi_{\text{garbage}} := \liminf_{m\to\infty} -\frac{1}{m}\log(2P_e^{\text{stale}}(m;\theta^{\text{bad}})) < \xi_{\text{L}}(\theta^{\text{bad}})$. Then:*

(i) ***Full-cycle.*** *If $\mathcal{R}_O$ contains a directed cycle enabling $\text{Dec}_{\text{L}} \leftrightarrows \text{Dec}_{\text{N}}$ and the observer monitors regimes with cost $C_{\text{monitor}} > 0$ and exponentially small misclassification probability, then there exists a policy achieving*

$$\xi_{\text{eff}} \;\ge\; (1 - C_{\text{monitor}})\Big(f\,\xi_{\text{N}}(\theta^{\text{good}}) + (1-f)\,\xi_{\text{L}}(\theta^{\text{bad}}) \Big).$$

(ii) ***Expand-only.*** *If $\mathcal{R}_O$ is expand-only with an absorbing linear state (no contraction edge $\text{Dec}_{\text{L}} \to \text{Dec}_{\text{N}}$), and if $\mathbb{P}(\theta_t = \theta^{\text{bad}}) > 0$, then any policy that avoids using $\text{Dec}_{\text{N}}^{\text{stale}}$ on bad-regime episodes must eventually remain in Dec_{L}, and hence*

$$\xi_{\text{eff}} = \xi_{\text{L}}(\theta^{\text{bad}}) \quad \textit{(permanently, after a transient).}$$

(iii) ***Fixed collective decoding without monitoring.*** *If the observer applies $\text{Dec}_{\text{N}}^{\text{stale}}$ in every episode without monitoring, then*

$$\xi_{\text{eff}} = f\,\xi_{\text{N}}(\theta^{\text{good}}) + (1-f)\,\xi_{\text{garbage}},$$

which can be arbitrarily small and may be negative if $P_e^{\text{stale}}(m;\theta^{\text{bad}}) > 1/2$ for large m.

Remark 5 (Interpretation)**.** Result A establishes that long-run redundancy is not determined solely by within-episode access, calibration, and memory, but also by whether the observer can *complete a revision cycle* after a period of operating under a robust (expanded) decoding assumption. The key invariant is topological: the existence of a directed cycle in $\mathcal{R}_O$ separates observers that can recover high-exponent collective decoding from observers that irreversibly collapse to the linear baseline.

9.3 Result B: inverted sophistication below a critical coherence fraction

Result B identifies a regime in which a collective measurement—despite being strictly more powerful under ideal conditions—yields higher error than product decoding when deployed without coherence monitoring. The mechanism is a mismatch between the measurement design (which presupposes inter-fragment coherence) and the actual regime (which may lack it).

Theorem 5 (Inverted sophistication threshold for unmonitored collective decoding)**.** *In the setting of Theorem 4, define*

$$f^\star := \frac{\xi_{\mathrm{L}}(\theta^{\mathrm{bad}}) - \xi_{\mathrm{garbage}}}{\xi_{\mathrm{N}}(\theta^{\mathrm{good}}) - \xi_{\mathrm{garbage}}} \;\in\; (0,1),$$

assuming $\xi_{\mathrm{garbage}} < \xi_{\mathrm{L}}(\theta^{\mathrm{bad}}) < \xi_{\mathrm{N}}(\theta^{\mathrm{good}})$*. If the coherence fraction satisfies* $f < f^\star$*, then the unmonitored stale collective decoder is strictly worse than optimal product decoding in the sense that*

$$\xi_{\mathrm{eff}}(\mathrm{Dec}_{\mathrm{N}}^{\mathrm{stale}}) < \xi_{\mathrm{eff}}(\mathrm{Dec}_{\mathrm{L}}^{\star}) = \xi_{\mathrm{L}}(\theta^{\mathrm{bad}}),$$

even though collective decoding is optimal in the good regime.

Proof. Under unmonitored stale collective decoding, $\xi_{\mathrm{eff}} = f\,\xi_{\mathrm{N}}(\theta^{\mathrm{good}}) + (1-f)\,\xi_{\mathrm{garbage}}$ (Theorem 4(iii)). Optimal product decoding achieves exponent $\xi_{\mathrm{L}}(\theta^{\mathrm{bad}})$ in both regimes by robustness (assumption). Rearranging the inequality $f\,\xi_{\mathrm{N}}(\theta^{\mathrm{good}}) + (1-f)\,\xi_{\mathrm{garbage}} < \xi_{\mathrm{L}}(\theta^{\mathrm{bad}})$ yields $f < f^\star$. □

Remark 6 (Per-episode versus time-averaged thresholds)**.** Theorem 5 characterizes the *time-averaged* effective exponent ξ_{eff}, yielding a threshold $f^\star$ that depends on the per-fragment exponents but not on the fragment count m. A complementary *per-episode* analysis considers the single-episode error probability $P_e^{\mathrm{un}}(m) = f\,P_e^N(m) + (1-f)/2$ and defines $f^*_{\mathrm{episode}}(m) := (P_e^L(m) - 1/2)/(P_e^N(m) - 1/2)$. Since $P_e^N(m) \to 0$ exponentially faster than $P_e^L(m)$, the per-episode threshold satisfies $f^*_{\mathrm{episode}}(m) \to 1$ as $m \to \infty$, with $1 - f^*_{\mathrm{episode}} \sim \exp(-\xi_L\, m)$. Thus the per-episode formulation predicts that for any fixed $f < 1$, there exists m^* beyond which *every individual episode* of unmonitored collective decoding has higher error than product decoding.

Remark 7 (Physical meaning of f and ξ_{garbage})**.** In a laboratory realization, f can be realized either as a *time fraction* of coherent episodes (interspersed with decohered episodes) or as an effective coherence parameter induced by storing an m-fragment register for time τ_O before a collective measurement. A simple model is an observer-memory dephasing channel with $f = e^{-\gamma\tau_O}$. The "garbage" exponent ξ_{garbage} is defined operationally as the error exponent achieved by a collective POVM optimized for a coherent register when applied to a decohered one; it quantifies the performance degradation incurred by unmonitored sophistication.

9.4 Robustness of inverted sophistication: observer-side versus system-side decoherence

The inverted sophistication threshold $f^\star$ depends on the physical mechanism by which coherence is lost. We distinguish two classes and verify that the qualitative prediction is robust to the choice of decoherence model, while its quantitative strength is not.

Definition 9 (Observer-side versus system-side decoherence)**.** Let $\{|\phi_k^{(x)}\rangle\}$ denote the conditional fragment states in an SBS witness experiment.

(i) *Observer-side decoherence.* The fragment states remain pure, but the observer's measurement apparatus fails to maintain the inter-fragment coherence required by the collective POVM. With probability $1-f$, the collective measurement returns an uninformative outcome; the product measurement, which acts on each fragment independently, is unaffected.

(ii) *System-side decoherence.* The fragment states themselves degrade via a depolarizing channel: $\sigma_k^{(x)} = f\,|\phi_k^{(x)}\rangle\langle\phi_k^{(x)}| + (1-f)\,I/2$. Both collective and product measurements operate on the same mixed-state register and are degraded symmetrically.

We analyze three concrete instantiations of the decoherence mechanism:

(a) **Binary episode-level (observer-side).** With probability f the collective POVM succeeds (error P_e^N); with probability $1-f$ it returns a random outcome ($P_e = 1/2$). The per-episode error is $P_e^{\mathrm{un}} = f\,P_e^N + (1-f)/2$. The per-episode critical coherence fraction satisfies $f^*_{\mathrm{episode}}(m) \to 1$ exponentially as $m \to \infty$, with $1 - f^*_{\mathrm{episode}} \sim \exp(-\xi_L\, m)$.

(b) **Continuous exponent scaling (observer-side).** Partial decoherence reduces the effective collective exponent to $f\,\xi_N$. The time-averaged critical coherence fraction is $f^\star = \xi_L/\xi_N$. For pure-state records where $\xi_N = 2\,\xi_L$, this gives $f^\star = 1/2$ independent of m.

(c) **Fragment depolarization (system-side).** Both measurement classes operate on the depolarized states $\sigma_k^{(x)}$. If the collective decoder is re-optimized for these mixed-state records, it cannot perform worse than the best product strategy because product POVMs are a subset of joint POVMs on the same register. Accordingly, the "inverted sophistication" effect is not expected for *matched* system-side decoherence models; it is a phenomenon of observer-side decoherence and/or model mismatch (a stale collective POVM applied outside its validity regime).

Remark 8 (Physical interpretation)**.** The absence of inversion under system-side decoherence and its presence under observer-side decoherence is consistent with the observer-quality thesis: the inverted sophistication effect is a property of the *observer's measurement apparatus*, not of the environment's state. Product measurements are inherently robust to apparatus decoherence because they require no inter-fragment quantum coherence. Collective measurements exploit such coherence when it is available but suffer catastrophically when it is absent and undetected. This asymmetry is precisely what the revision graph $\mathcal{R}_O$ is designed to manage: a full-cycle topology enables the observer to detect coherence loss and fall back to the robust product strategy.

9.5 Result C: pointer accessibility and observer-relative einselection

Result C formalizes a stage-dependent *resolution limit*: for finite fragment budgets, the DLN stage determines which pointer distinctions are operationally recoverable from environment fragments.

Definition 10 (Stage-conditional pointer accessibility)**.** Fix a fragment budget m and target error $\delta \in (0, 1/2)$. Let Π be a candidate pointer partition (a coarse-graining of $\mathcal{X}$) and let $\{\rho_A^{(\pi)}\}$ be the induced conditional fragment states for $\pi \in \Pi$. We say that Π is (m,δ)-*accessible* to an observer in decoder class q if there exists a decoder in class q acting on m fragments whose worst-case error over $\pi \in \Pi$ is at most δ. Let $\mathfrak{P}(q; m, \delta)$ denote the set of all (m,δ)-accessible pointer partitions under class q.

Proposition 2 (Pointer accessibility gap across DLN stages)**.** *There exist SBS witness experiments and target accuracies* (m, δ) *for which*

$$\mathfrak{P}(q_{\mathrm{L}}; m, \delta) \subsetneq \mathfrak{P}(q_{\mathrm{N}}; m, \delta).$$

In particular, for pure-state records with overlap $c \in (0, 1)$*, any pointer distinction that requires an exponent exceeding* $-\log c$ *is inaccessible under product decoding within budget* m *but can be accessible under collective decoding because the achievable exponent can reach* $-\log(c^2)$ *(Cor. 1).*

Proof sketch. The inclusion $\subseteq$ holds because any product decoder is a special case of a collective decoder on the same register. Strictness follows by choosing a record model where the error target δ and budget m satisfy $m(-\log(c^2)) \geq \log(1/(2\delta))$ but $m(-\log c) < \log(1/(2\delta))$. Then collective decoding can meet the target while product decoding cannot, by Cor. 1. □

Corollary 2 (Observer-relative einselection under fragment constraints)**.** *If multiple approximately stable pointer partitions are supported by the same environment but differ in required fragment budget, then the* operationally objective *pointer partition selected by an observer depends on* (q_O, τ_O) *through the accessibility sets* $\mathfrak{P}(q_O; m, \delta)$*. In this sense, "einselection" can be formulated as a constrained optimization over pointer partitions, and designing the observer (its DLN stage and revision topology) selects which pointer distinctions are controllably accessible from a given fragment budget.*

9.6 Numerical verification in the central-spin model

We verify Results A–C in the central-spin pure-dephasing model of Sec. 7 with $N = 200$ environment qubits, couplings $g_k \sim \mathrm{Uniform}(0.8, 1.2)$, interaction time $t = 0.5$, and monitoring fraction $C_{\mathrm{monitor}} = 0.10$. Unless otherwise stated we use a fixed random seed (seed = 2) for reproducibility. The accompanying scripts in `scripts/` reproduce all numerical values and figures reported here.

Result A (dynamical redundancy). At $m = 25$ fragments, the per-fragment collective and local-Helstrom exponents are $\xi_N/m = 1.915$ and $\xi_L/m = 0.958$, confirming the ratio $\xi_N/\xi_L = 2.00$ (Cor. 1). To quantify the impact of unmonitored failures, we report the ensemble exponent ξ_{ens} (Remark after Eq. (8)). At coherence fraction $f = 0.8$, with monitoring fraction $C_{\mathrm{monitor}} = 0.10$ (implemented as $m_{\mathrm{diag}} = \lceil 0.10\, m \rceil = 3$ diagnostic fragments and $m_{\mathrm{dec}} = 22$ decoding fragments), the ensemble exponents are:

$\xi_{\mathrm{ens}}^{\mathrm{Full}} = 0.912$ (full-cycle with monitoring), $\xi_{\mathrm{ens}}^{L} = 0.958$ (fixed local decoding), $\xi_{\mathrm{ens}}^{\mathrm{un}} = 0.064$ (unmonito

The unmonitored collective decoder collapses toward zero because bad-regime episodes contribute an m-independent error floor, so the average error does not decay exponentially with m.

Result B (inverted sophistication). For the binary episode-level observer-side decoherence model (Sec. 9.4(a)), the per-episode critical coherence fraction $f^*_{\mathrm{episode}}(m)$ satisfies $f^*_{\mathrm{episode}}(5) = 0.993$, $f^*_{\mathrm{episode}}(10) = 0.9999$, and $f^*_{\mathrm{episode}}(m) \to 1$ rapidly for $m \gtrsim 15$. For the continuous scaling model (Sec. 9.4(b)) with the pure-state factor-of-two gap, the coherence threshold is $f^\star = 1/2$. For system-side depolarization (Sec. 9.4(c)), re-optimized collective decoding is never worse than product decoding, and our numerical checks show no inversion over the tested $f \in (0, 1]$ and $m \in \{5, 10, 15, 20, 25\}$.

Result C (pointer accessibility). For target error $\delta = 10^{-3}$ and times sampled uniformly on $t \in [0, 2]$ (1001 grid points), we compute the set of times where q_{N} achieves the decoding target within budget m but q_{L} does not. The resulting resolution-gap fractions are 19.48% at $m = 8$, 17.78% at $m = 12$, 9.99% at $m = 25$, and 2.997% at $m = 50$, consistent with Proposition 2.

10 Discussion: tightness, limitations, and relation to prior work

We highlight three points relevant for interpreting the bounds and comparing to prior work:

- **Mathematical depth.** The main redundancy bound employs Chernoff-optimal decoding [**? ?**] and explicitly incorporates calibration via a CPTP contraction theorem (Thm. 2) and ε-SBS perturbations (Thm. 3). The local-versus-collective decoding gap (Cor. 1) provides a quantitative link between temporal observer constraints and reduced error exponents.
- **Robustness of dynamical predictions.** The inverted sophistication threshold (Result B) is robust to the choice of decoherence model in the following precise sense: under any observer-side decoherence mechanism, there exists a critical coherence fraction below which unmonitored collective decoding is strictly inferior to product decoding. The quantitative threshold ranges from $f^\star = 1/2$ (continuous model) to $f^\star \to 1$ (binary model). Under system-side decoherence (fragment depolarization), no inversion occurs, confirming that the effect is a property of the observer's measurement apparatus rather than of the environment's state (Sec. 9.4).
- **Prior work.** We compare to hazy environments [**?**], photon redundancy [**? ?**], generic objectivity [**? ?**], and accessible-information approaches [**?**].
- **Concrete physical verification.** Section 7 provides a fully worked open-system example, and Sec. 9.6 verifies all three dynamical predictions numerically in the same model.
- **DLN connection.** The DLN representational framework is summarized in Sec. 8; the quantitative claims of this paper are self-contained.

11 Conclusion

Observer modeling remains a structural gap in much of the QD/SBS literature. By treating the observer as a resource-constrained agent with explicit access, calibration, and temporal horizons, we obtain quantitative, decoder-level statements about redundancy and objectivity that remain stable under ε-SBS trace-distance approximations. The central-spin worked example demonstrates how these parameters can be computed from physical couplings and instrument noise, enabling model-to-experiment predictions.

The three dynamical predictions of Sec. 9 extend the static analysis to non-stationary observation protocols. The revision-graph topology $\mathcal{R}_O$ determines whether an observer can maintain a positive long-run Chernoff exponent under fluctuating coherence (Result A), predicts a regime in which unmonitored collective decoding is strictly inferior to product decoding (Result B), and identifies stage-dependent resolution limits on pointer accessibility (Result C). The robustness analysis of Sec. 9.4 establishes that the inverted sophistication effect is specific to observer-side decoherence, providing a testable distinction between apparatus limitations and environment-state degradation.

These results suggest that the observer's information-processing topology—not merely its access, calibration, or temporal horizon—is a first-class resource in quantum Darwinism. Characterizing and

controlling this topology may enable the design of measurement protocols that achieve near-optimal redundancy under realistic, non-stationary conditions.